UNIVERSITY OF WESTMINSTER

INFORMATICS INSTITUTE OF TECHNOLOGY

4BUIS015C.1 Database Design and Implementation

COURSEWORK

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Contents

ACKNOWLEDGMENT 2
List of figures 4
1.ER DIAGRAM 5
Assumptions 5
2. Entity, Keys and relationship table 7
3. Creating tables and inserting data 9
1.W2120869_Customer 9
2.W31120869_Department 10
3.W2120869_Employee 11
4.W2120869_Customer_Employee 12
5.W2120869_Order_Table 13
6.W2120869_Commission 14
7.W2120869_Product 15
8.W2120869_Item 16
4.Query and output table 17
5.Query outputs 20

List of figures

Figure 1 ERD 6
Figure 2 9
Figure 3 9
Figure 4 10
Figure 5 10
Figure 6 11
Figure 7 11
Figure 8 12
Figure 9 12
Figure 10 13
Figure 11 13
Figure 12 14
Figure 13 14
Figure 14 15
Figure 15 15
Figure 16 16
Figure 17 16
Figure 18 Query 1 20
Figure 19 Query 2 20
Figure 20 Query 3 21
Figure 21 Query 4 21
Figure 22 Query 5 22
Figure 23 Query 6 22
Figure 24 Query 7 23
Figure 25 Query 8 23
Figure 26 Query 9 24
Figure 27 Query 10 24

1.ER DIAGRAM

Assumptions

- W_2120869_Customer_Employee table shows the interaction between customers and employees which is a result after solving the many to many relationship between customer table and employee table.
- One interaction can have only one employee.
- Per item can belong to only one product category.
- Category in W2120869_Product table refers to the sport category that the product belongs.
- Employees can work on 0 to one department as some employees may work on logistics and one employee can work on only one department.
- Per item is unique. So one item can belong to only one order.

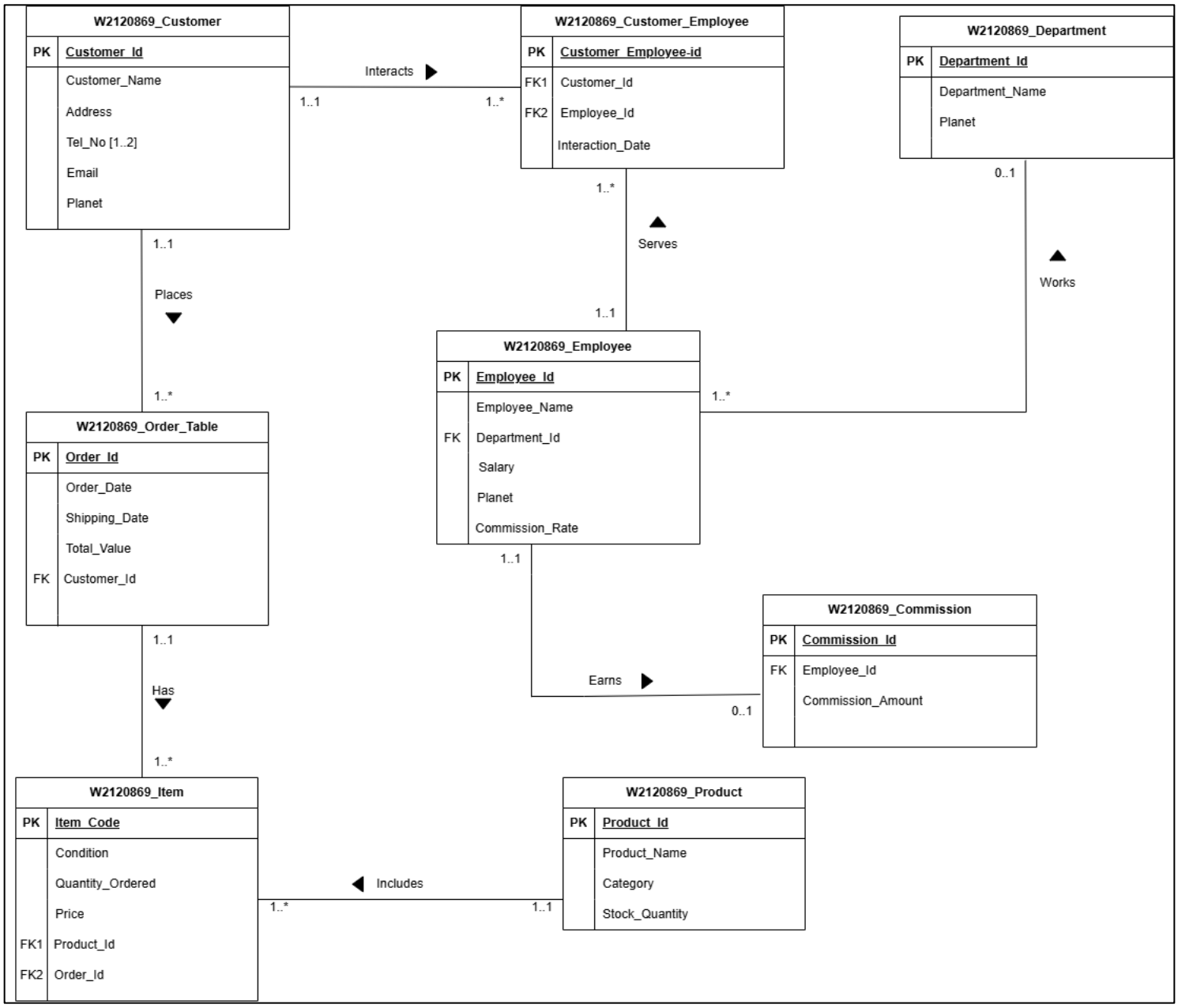
W2120869_Customer
PK Customer_Id
Customer_Name
Address
Tel_No [1..2]
Email
Planet
Interacts
1..1
1..*
W2120869_Customer_Employee
PK Customer_Employee-id
FK1 Customer_Id
FK2 Employee_Id
Interaction_Date
W2120869_Department
PK Department_Id
Department_Name
Planet
0..1
1..*
Serves
1..1
Works
1..1
Places
1..*
W2120869_Employee
PK Employee_Id
Employee_Name
FK Department_Id
Salary
Planet
Commission_Rate
1..*
W2120869_Order_Table
PK Order_Id
Order_Date
Shipping_Date
Total_Value
FK Customer_Id
1..1
Earns
0..1
W2120869_Commission
PK Commission_Id
FK Employee_Id
Commission_Amount
1..1
Has
1..*
W2120869_Item
PK Item_Code
Condition
Quantity_Ordered
Price
FK1 Product_Id
FK2 Order_Id
Includes
1..*
1..1
W2120869_Product
PK Product_Id
Product_Name
Category
Stock_Quantity

Figure 1 ERD

2. Entity, Keys and relationship table

Table 1 Entity, keys & relationships

Entity	**Primary Key PK**	**Foreign Key FK**	**Rationale for PK**
W2120869_Customer	Customer_Id	N/A	Customer_Id uniquely identifies each customer in the system, necessary for tracking orders and interactions.
W2120869_Employee	Employee_Id	Department_Id	Employee_Id uniquely identifies each employee for payroll, commissions, and departmental assignments.
W2120869_Customer_Employee	Customer_Employee_Id	Customer_Id Employee_Id	Customer_Employee_Id tracks specific customer employee interactions.
W2120869_Department	Department_Id	N/A	Department_Id ensures each department is uniquely identifiable as they may exists across planets.
W2120869_Order_Table	Order_Id	Customer_Id	Order_Id is essential to track customer orders, shipping details and total values.
W2120869_Commission	Commission_Id	N/A	Commission_Id is unique to track commission payments tied to employees.
W2120869_Product	Product_Id	N/A	Product_Id uniquely identifies each product to manage stock and link to related items.

W2120869_Item	Item_Code	Product_Id Order_Id	Item_Code is a unique identifier to track individual inventory items and their attributes such as category.

3. Creating tables and inserting data

1.W2120869_Customer

Run SQL query/queries on database sportsgalactic:

```sql
CREATE TABLE W2120869_Customer
(Customer_Id INT PRIMARY KEY NOT NULL,
 Customer_Name varchar(50) NOT NULL,
 Address varchar(100) NOT NULL,
 Tel_No_1 INT(10) NOT NULL,
 Tel_No_2 INT(10) NULL,
 Email varchar(50)NULL,
 Planet varchar(10) NOT NULL);
```

Figure 2

```sql
INSERT INTO W2120869_Customer (Customer_Id, Customer_Name, Address, Tel_No_1, Tel_No_2, Email, Planet)
VALUES
  (1, 'John Doe', '123 Main St,Earth', 1234567890, 9876543210, 'johndoe@email.com', 'Earth'),
  (2, 'Jane Smith', '456 Elm St,Jupiter', 9876543210, NULL, 'janesmith@email.com', 'Jupiter'),
  (3, 'Michael Johnson', '789 Oak Ave,Earth', 1234567890, 5432109876, 'michaeljohnson@email.com', 'Earth'),
  (4, 'Emily Davis', '101 Pine Rd, Mercury', 9876543210, NULL, 'emilydavis@email.com', 'Mercury'),
  (5, 'David Wilson', '234 Cedar Ln,Venus', 1234567890, 6789012345, 'davidwilson@email.com', 'Venus'),
  (6, 'Olivia Brown', '567 Maple Ave,Jupiter', 9876543210, NULL, 'oliviabrown@email.com', 'Jupiter'),
  (7, 'William Taylor', '890 Birch St,Venus', 1234567890, 3210987654, 'williamtaylor@email.com', 'Venus'),
  (8, 'Sophia Anderson', '111 Spruce Ct, Saturn', 9876543210, NULL, 'sophiaanderson@email.com', 'Saturn'),
  (9, 'James Thomas', '222 Fir Ln,Earth', 1234567890, 7654321098, 'jamesthomas@email.com', 'Earth'),
  (10, 'Ava Lee', '333 Cypress Rd,Venus', 9876543210, NULL, 'avalee@email.com', 'Venus');
```

Figure 3

2.W31120869_Department

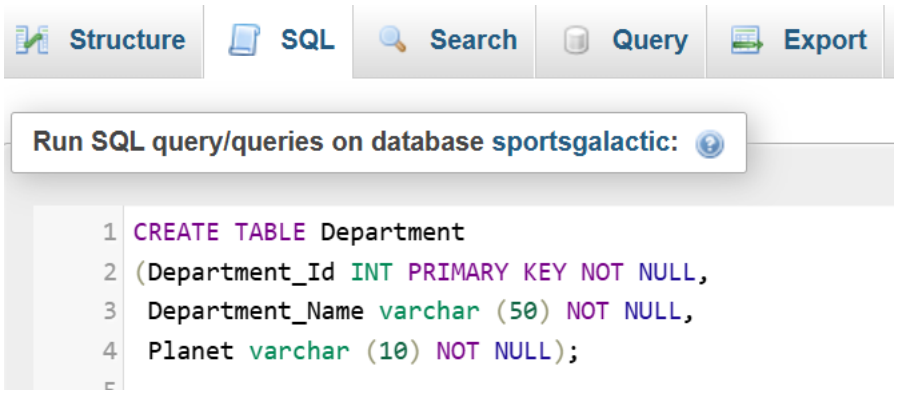

Figure 4

```
 INSERT INTO Department (Department_Id, Department_Name, Planet)
VALUES
  (011, 'Sales', 'Earth'),
  (012, 'Human Resources', 'Mars'),
  (013, 'Planetary Logistics', 'Venus'),
  (014, 'Accountant', 'Mercury'),
  (015, 'Marketing', 'Jupiter'),
  (016, 'Customer Service', 'Saturn'),
  (017, 'IT', 'Uranus'),
  (018, 'Finance', 'Neptune'),
  (019, 'Research and Development', 'Pluto'),
  (110, 'Legal', 'Earth');
```

Figure 5

3.W2120869_Employee

Structure | SQL | Search | Query | Export | Import | Operations

Run SQL query/queries on database sportsgalactic:

```sql
CREATE TABLE W2120869_Employee (
    Employee_Id INT PRIMARY KEY NOT NULL,
    Employee_Name VARCHAR(50) NOT NULL,
    Salary DECIMAL(6,2) NOT NULL,
    Planet VARCHAR(10) NOT NULL,
    Commission_Rate DECIMAL(5,2) NOT NULL,
    Department_Id INT NOT NULL,
    FOREIGN KEY (Department_Id) REFERENCES W2120869_Department(Department_Id)
);
```

Figure 6

```sql
INSERT INTO W2120869_Employee (Employee_Id, Employee_Name, Salary, Planet, Commission_Rate, Department_Id)
VALUES
(111, 'Anne', 5000.00, 'Earth', 0.10, 011),
(112, 'James', 6000.00, 'Mars', 0.20, 012),
(113, 'William', 4500.00, 'Venus', 0.10, 013),
(114, 'Emily', 5500.00, 'Mercury', 0.20, 014),
(115, 'Jane', 6500.00, 'Jupiter', 0.10, 015),
(116, 'Hector', 4000.00, 'Saturn', 0.20, 016),
(117, 'Taylor', 5200.00, 'Uranus', 0.10, 017),
(118, 'Sabrina', 6200.00, 'Neptune', 0.20, 018),
(119, 'Henry', 4800.00, 'Pluto', 0.10, 019),
(1110, 'Max', 5800.00, 'Earth', 0.20, 110);
```

Figure 7

4.W2120869_Customer_Employee

Run SQL query/queries on database sportsgalactic:

```sql
CREATE TABLE W2120869_Customer_Employee (
    Customer_Employee_Id VARCHAR(5) NOT NULL,
    Customer_Id INT NOT NULL,
    Employee_Id INT NOT NULL,
    Interaction_Date DATE NOT NULL,
    FOREIGN KEY (Customer_Id) REFERENCES W2120869_Customer(Customer_Id),
    FOREIGN KEY (Employee_Id) REFERENCES W2120869_Employee(Employee_Id)
);
```

Figure 8

```sql
INSERT INTO W2120869_Customer_Employee (Customer_Employee_Id, Customer_Id, Employee_Id, Interaction_Date)
VALUES
('INT1', 1, 111, '2023-11-20'),
('INT2', 2, 112, '2023-11-21'),
('INT3', 3, 113, '2023-11-22'),
('INT4', 4, 114, '2023-11-23'),
('INT5', 5, 115, '2023-11-24'),
('INT6', 6, 116, '2023-11-25'),
('INT7', 7, 117, '2023-11-26'),
('INT8', 8, 118, '2023-11-27'),
('INT9', 9, 119, '2023-11-28'),
('INT10', 10, 1110, '2023-11-29');
```

Figure 9

5.W2120869_Order_Table

```sql
CREATE TABLE Order_Table
(Order_Id varchar(5) PRIMARY KEY NOT NULL,
 Order_Date DATE NOT NULL,
 Shipping_Date DATE NOT NULL,
 Total_Value DECIMAL(10,2) NOT NULL,
 Customer_Id INT NOT NULL,
 FOREIGN KEY(Customer_Id) REFERENCES w2120869_customer(Customer_Id));
```

Figure 10

```sql
 INSERT INTO Order_Table (Order_Id, Order_Date, Shipping_Date, Total_Value, Customer_Id)
VALUES
  ('ORD1', '2023-11-20', '2023-11-25', 1000.00, 1),
  ('ORD2', '2023-11-21', '2023-11-26', 1500.00, 2),
  ('ORD3', '2023-11-22', '2023-11-27', 800.00, 3),
  ('ORD4', '2023-11-23', '2023-11-28', 2000.00, 4),
  ('ORD5', '2023-11-24', '2023-11-29', 1200.00, 5),
  ('ORD6', '2023-11-25', '2023-11-30', 900.00, 6),
  ('ORD7', '2023-11-26', '2023-12-01', 1800.00, 7),
  ('ORD8', '2023-11-27', '2023-12-02', 700.00, 8),
  ('ORD9', '2023-11-28', '2023-12-03', 2500.00, 9),
  ('ORD10', '2023-11-29', '2023-12-04', 1300.00, 10);
```

Figure 11

6.W2120869_Commission

```sql
CREATE TABLE W2120869_commission
(Commission_Id varchar(5) NOT NULL,
 Employee_Id INT NOT NULL,
 Commission_Amount DECIMAL(5,2) NOT NULL,
 FOREIGN KEY (Employee_Id) REFERENCES W2120869_Employee(Employee_Id));
```

Figure 12

```sql
 INSERT INTO W2120869_commission (Commission_Id, Employee_Id, Commission_Amount)
VALUES
('COM1', 111, 500.00),
('COM2', 112, 1200.00),
('COM3', 113, 450.00),
('COM4', 114, 1100.00),
('COM5', 115, 650.00),
('COM6', 116, 400.00),
('COM7', 117, 520.00),
('COM8', 118, 1240.00),
('COM9', 119, 480.00),
('COM10', 1110, 1160.00);
```

Figure 13

7.W2120869_Product

```sql
CREATE TABLE W2120869_Product
(Product_Id varchar(5) PRIMARY KEY NOT  NULL,
 Product_Name varchar(50) NOT NULL,
 Category varchar(10) NOT NULL,
 Stock_Quantity INT NOT NULL);
```

Figure 14

```sql
INSERT INTO W2120869_product (Product_Id, Product_Name, Category, Stock_Quantity)
VALUES
  ('PRD1', 'Hoverball', 'Sports', 100),
  ('PRD2', 'Astro Wrestling Gear', 'Sports', 50),
  ('PRD3', 'Anti-Gravity Racing Suit', 'Sports', 20),
  ('PRD4', 'Zero-G Basketball', 'Sports', 80),
  ('PRD5', 'Alien Frisbee', 'Sports', 120),
  ('PRD6', 'Nebula Net', 'Sports', 60),
  ('PRD7', 'Cosmic Cleats', 'Sports', 90),
  ('PRD8', 'Quantum Quaffle', 'Sports', 40),
  ('PRD9', 'Stellar Soccer Ball', 'Sports', 150),
  ('PRD10', 'Galactic Golf Club', 'Sports', 70);
```

Figure 15

8.W2120869_Item

```sql
CREATE TABLE W2120869_Item (
    Item_Code VARCHAR(5) NOT NULL PRIMARY KEY,
    Condition_ VARCHAR(10) NOT NULL,
    Quantity_Ordered INT NOT NULL,
    Price DECIMAL(10,2) NOT NULL,
    Product_Id VARCHAR(5) NOT NULL,
    Order_Id VARCHAR(5) NOT NULL,
    FOREIGN KEY (Product_Id) REFERENCES W2120869_product(Product_Id),
    FOREIGN KEY (Order_Id) REFERENCES W2120869_order_table(Order_Id)
);
```

Figure 16

```sql
INSERT INTO W2120869_Item (Item_Code, Condition_, Quantity_Ordered, Price, Product_Id, Order_Id)
VALUES
('ITM1', 'New', 2, 100.00, 'PRD1', 'ORD1'),
('ITM2', 'Used', 1, 50.00, 'PRD2', 'ORD2'),
('ITM3', 'Refurbished', 3, 80.00, 'PRD3', 'ORD3'),
('ITM4', 'New', 4, 200.00, 'PRD4', 'ORD4'),
('ITM5', 'Used', 2, 120.00, 'PRD5', 'ORD5'),
('ITM6', 'Refurbished', 3, 90.00, 'PRD6', 'ORD6'),
('ITM7', 'New', 2, 180.00, 'PRD7', 'ORD7'),
('ITM8', 'Used', 1, 70.00, 'PRD8', 'ORD8'),
('ITM9', 'Refurbished', 3, 250.00, 'PRD9', 'ORD9'),
('ITM10', 'New', 2, 130.00, 'PRD10', 'ORD10');
```

Figure 17

4.Query and output table

Table 2 Query & Output

Query No.	Purpose	Tables involved	SQL	Correct execution
1	List all the products that are low in stock(<50)	W2120869_Product	SELECT Product_Id,Product_Name FROM w2120869_product WHERE Stock_Quantity <50;	Yes
2	Find the top 10 customers with the highest total purchase value	W2120869_Customer W2120869_Order_Table	SELECT c.Customer_Id, c.Customer_Name, SUM(o.Total_Value) AS TotalPurchase FROM w2120869_customer c JOIN w2120869_order_table o ON c.Customer_Id = o.Customer_Id GROUP BY c.Customer_Id, c.Customer_Name ORDER BY TotalPurchase DESC;	Yes
3	Finding the most popular product	W2120869_Product W2120869_Item	SELECT p.Product_Id, p.Product_Name, COUNT(*) AS OrderCount FROM w2120869_product p JOIN w2120869_item i ON p.Product_Id = i.Product_Id GROUP BY p.Product_Id, p.Product_Name ORDER BY OrderCount DESC;	Yes

4	List the customer names and orderno's for all orders placed on 26th Nov 2023.	W2120869_Custtomer W2120869_Order_Table	Select DISTINCT Customer_Name,Order_Id from w2120869_customer c,w2120869_order_table O where c.Customer_Id = O.Customer_Id AND Order_Date = '2023-11-26';	Yes
5	Identify slow-moving products	W2120869_Product W2120869_Item	SELECT p.Product_Id, p.Product_Name, SUM(i.Quantity_Ordered) AS TotalQuantitySold FROM w2120869_product p JOIN w2120869_item i ON p.Product_Id = i.Product_Id GROUP BY p.Product_Id, p.Product_Name HAVING SUM(i.Quantity_Ordered) < 10;	Yes
6	Optimize inventory levels	W2120869_Item	SELECT i.Item_Code, SUM(i.Quantity_Ordered) AS Total_Quantity_Sold FROM W2120869_Item i GROUP BY i.Item_Code;	Yes

7	Analyze product performance by planet	W2120869_Product W2120869_Item W2120869_Order_Table W2120869_Customer	SELECT p.Product_Name, c.Planet, SUM(i.Quantity_Ordered) AS Total_Quantity_Sold FROM W2120869_Product p JOIN W2120869_Item i ON p.Product_ID = i.Product_ID JOIN W2120869_Order_Table o ON i.Order_ID = o.Order_ID JOIN W2120869_Customer c ON o.Customer_ID = c.Customer_ID GROUP BY p.Product_Name, c.Planet;	Yes
8	Show tototal sales value grouped by each customer	W2120869_Customer W2120869_Order_Table	SELECT c.Customer_Name,SUM(o.Total_Value) AS Total_Sales_Values_ FROM w2120869_order_table o JOIN w2120869_customer c ON o.Customer_Id=c.Customer_Id GROUP BY c.Customer_Id,c.Customer_Name;	Yes
9	Analyze product performance by planet	W2120869_Product W2120869_Item W2120869_Order_Table W2120869_Customer	SELECT p.Product_Name, c.Planet, SUM(i.Quantity_Ordered) AS Total_Quantity_Sold, AVG(i.Price) AS Avg_Price FROM W2120869_Product p JOIN W2120869_Item i ON p.Product_Id= i.Product_Id JOIN W2120869_Order_Table o ON i.Order_Id = o.Order_Id JOIN W2120869_Customer c ON o.Customer_Id = c.Customer_Id GROUP BY p.Product_Name, c.Planet;	Yes
10	Commission earned by employees	W2120869_Employee W2120869_Commission	SELECT Employee_Name, Commission_Amount FROM w2120869_employee JOIN w2120869_commission ON w2120869_employee.Employee_Id = w2120869_commission.Employee_Id;	Yes

5.Query outputs

1.

Extra options

				Product_Id	Product_Name
☐	Edit	Copy	Delete	PRD3	Anti-Gravity Racing Suit
☐	Edit	Copy	Delete	PRD8	Quantum Quaffle

Check all *With selected:* Edit Copy Delete Export

Figure 18 Query 1

2.

Customer_Id	Customer_Name	TotalPurchase 1
9	James Thomas	2500.00
4	Emily Davis	2000.00
7	William Taylor	1800.00
2	Jane Smith	1500.00
10	Ava Lee	1300.00
5	David Wilson	1200.00
1	John Doe	1000.00
6	Olivia Brown	900.00
3	Michael Johnson	800.00
8	Sophia Anderson	700.00

Figure 19 Query 2

3.

Product_Id	Product_Name	OrderCount 1
PRD8	Quantum Quaffle	1
PRD7	Cosmic Cleats	1
PRD6	Nebula Net	1
PRD5	Alien Frisbee	1
PRD4	Zero-G Basketball	1
PRD3	Anti-Gravity Racing Suit	1
PRD2	Astro Wrestling Gear	1
PRD10	Galactic Golf Club	1
PRD1	Hoverball	1
PRD9	Stellar Soccer Ball	1

Figure 20 Query 3

4.

Showing rows 0 - 0 (1 total, Query took 0.0029 seconds.)

```
Select DISTINCT Customer_Name,Order_Id from w2120869_customer c,w2120869_order_tabl
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

Customer_Name	Order_Id
Olivia Brown	ORD6

Figure 21 Query 4

5.

Product_Id	Product_Name	TotalQuantitySold
PRD1	Hoverball	2
PRD10	Galactic Golf Club	2
PRD2	Astro Wrestling Gear	1
PRD3	Anti-Gravity Racing Suit	3
PRD4	Zero-G Basketball	4
PRD5	Alien Frisbee	2
PRD6	Nebula Net	3
PRD7	Cosmic Cleats	2
PRD8	Quantum Quaffle	1
PRD9	Stellar Soccer Ball	3

Figure 22 Query 5

6.

←T→				Item_Code	Total_Quantity_Sold
☐	Edit	Copy	Delete	ITM1	2
☐	Edit	Copy	Delete	ITM10	2
☐	Edit	Copy	Delete	ITM2	1
☐	Edit	Copy	Delete	ITM3	3
☐	Edit	Copy	Delete	ITM4	4
☐	Edit	Copy	Delete	ITM5	2
☐	Edit	Copy	Delete	ITM6	3
☐	Edit	Copy	Delete	ITM7	2
☐	Edit	Copy	Delete	ITM8	1
☐	Edit	Copy	Delete	ITM9	3

Figure 23 Query 6

7.

Product_Name	Planet	Total_Quantity_Sold
Alien Frisbee	Venus	2
Anti-Gravity Racing Suit	Earth	3
Astro Wrestling Gear	Jupiter	1
Cosmic Cleats	Venus	2
Galactic Golf Club	Venus	2
Hoverball	Earth	2
Nebula Net	Jupiter	3
Quantum Quaffle	Saturn	1
Stellar Soccer Ball	Earth	3
Zero-G Basketball	Mercury	4

Figure 24 Query 7

8.

Customer_Name	Total_Sales_Values_
John Doe	1000.00
Jane Smith	1500.00
Michael Johnson	800.00
Emily Davis	2000.00
David Wilson	1200.00
Olivia Brown	900.00
William Taylor	1800.00
Sophia Anderson	700.00
James Thomas	2500.00
Ava Lee	1300.00

Figure 25 Query 8

9.

Product_Name	Planet	Total_Quantity_Sold	Avg_Price
Alien Frisbee	Venus	2	120.000000
Anti-Gravity Racing Suit	Earth	3	80.000000
Astro Wrestling Gear	Jupiter	1	50.000000
Cosmic Cleats	Venus	2	180.000000
Galactic Golf Club	Venus	2	130.000000
Hoverball	Earth	2	100.000000
Nebula Net	Jupiter	3	90.000000
Quantum Quaffle	Saturn	1	70.000000
Stellar Soccer Ball	Earth	3	250.000000
Zero-G Basketball	Mercury	4	200.000000

Figure 26 Query 9

10.

Employee_Name	Commission_Amount
Anne	500.00
James	999.99
William	450.00
Emily	999.99
Jane	650.00
Hector	400.00
Taylor	520.00
Sabrina	999.99
Henry	480.00
Max	999.99

Figure 27 Query 10